Baby Steps 1, 2, 3D

With the interest in 3D printing holding strong, we give you the lowdown on a web application that will help you take your first couple of steps into the rather complex world of 3D modeling

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The application in question here is 3DTin (www.3dtin.com), a web-based 3D editor for use in web browsers. Its clean layout gives you the best of both worlds by being as easy-to-use as MS Paint while providing some pretty high quality results. That's right, no experience whatsoever of 3D design is required.

Requirements and the layout

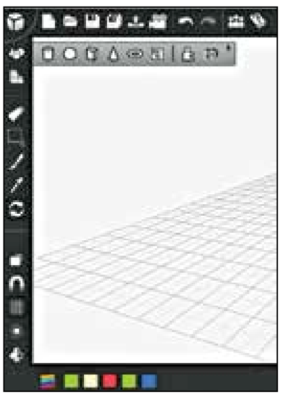
If you're accessing the app on Chrome or Firefox, no download or installation is required for this free-to-use 3D modeling tool. Since all sketches are stored on the cloud, you can easily access them from anywhere and share them with others via a unique URL. The service uses webGL so Internet Explorer isn't supported. And a compatible browser is all you really need.

The layout is, as we've said before, very clean with all your basic tools arranged on the left. Perpendicularly above the toolbar are the options to save, open and create new projects. A very interesting feature here is the 'Camera Animation' button, which allows you to rotate your project to view it in 3D and record the movements. The next part of the layout is the color bar located at the left bottom of the screen. This is also where you can store a pallet of colors with a color picker. The workspace is a 2D grid which serves as a flat base that works as a reference. Placed at the bottom-right corner are the camera controls.

Basic tools

Here's a basic overview of each of the 12 tools starting from the top of the toolbar and going to the bottom:

1. **Add Geometries:** Allows you to generate objects from an array of preset shapes. You can also create your own shapes using 'Make your own geometry' while 'Make geometry from 2D Shapes' lets you create 2D objects and convert them to 3D.
2. **Add Cube:** Generates perfect cubes that can be placed next to each other to draw other 3D objects.
3. **Erase:** Does what it says. Removes unwanted objects from your workspace.
4. **Select:** Allows the user to select multiple objects by holding down the left button and dragging the mouse to draw a rectangle. This is similar to the way you'd select multiple icons on your desktop.
5. **Color Change:** Allows the user to change the color of each individual object one by one.
6. **Color Picker:** Gives the user clarity about which color the desired object will be changed to if used before the 'Color Change' tool.
7. **View Rotate:** Allows the user to rotate the workspace through the X, Y and Z axes.
8. **Orthographic/Perspective:** Switches between a real, 3D representation of shapes and a simpler 2D representation of 3D objects.
9. **Snap to Grid:** On by default, it makes objects align themselves to the nearest intersection of the displayed grid on the 2D workspace. The grid can be shown/hidden by clicking the 'Grid on/off' option.
10. **Light options:** Allows you to light the object from various angles using a variety of sources.
11. **Symmetric Add:** Allows you to easily place a mirror image of an object anywhere in your workspace.
12. **Super Symmetry (premium feature):** Allows the user to draw a mirror image of any object in real time as well.



Easy-to-understand tools with a quick note that pops up when you hover over them

The Object toolbar

Once you generate an object, clicking on it brings up another toolbar that lets you to make further changes to the object. Using this toolbar you can:

- **Rotate**
- **Scale:** Extends the object symmetrically or through one or two of the axes.
- **Shear:** Keeps any two sides on the same plane while angling the other two sides. Imagine turning a rectangle into a parallelogram.
- **Flip:** Flips the object either vertically or horizontally.
- **Clone:** Creates a copy of the object.
- **Delete:** Gets rid of the object.
- **Group/Ungroup:** Appears when you select two or more objects to create one single object.

Let's build some 3D models!

As you begin to get the hang of 3DTin, you may find yourself spending a lot of time constructing shapes using cubes. Try to avoid this since the edit options normally